

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM AND LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/2-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime

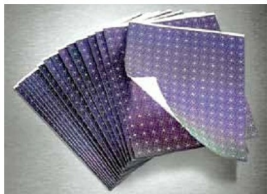


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Solar cells that use 95 per cent less silicon

In a big development that would considerably reduce solar-cell production costs, researchers at the University of Oslo, Norway, have developed silicon solar cells that are 20 times thinner than the existing ones. These solar cells use 95 per cent less silicon.



Thin solar cells (Courtesy: <http://www.alt-energy.info/>)

The researchers managed to trap the longer wavelengths—even when the silicon wafers were just 10 micrometres thick—by using microbeads that are very small plastic spheres, uniform in size. These microbeads created an almost perfect periodic pattern on the silicon that forced the sunlight to 'move sideways,' increasing the apparent thickness of silicon by 25 times.

Nanowire-based solar cells created

A team of Swedish and German researchers, led by Magnus Borgstrom from the Lund University, has made solar cells from tiny wires of the indium-phosphide (InP) semiconductor that cover only about 12 per cent surface of a device but have efficiencies as high as 13.8 per cent. The new cells measure about a square millimetre and contain about four million InP nanowires.

Talking about the project, Borgstrom said, "Our nanowires needed to be uniform, having a certain diameter and length in a certain pitch. From our first working p-n InP junctions, it has taken us four years to reach this result."

A smart pen that helps to write legibly and correctly

Today, people are drifting apart from the art of handwriting due to plethora of touch-based gadget options like smartphones, tablets and even laptops.



Smart pen from Lemstift (Courtesy: <http://www.dnsee.com/>)